

Deepak Mallampati

AI INFRASTRUCTURE ENGINEER - LLM TOOLING & OBSERVABILITY

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US-based (F-1 OPT) - open to EU relocation & authorization

PROFESSIONAL SUMMARY

AI infrastructure engineer with 4+ years building the systems LLMs run on - **tool integration** and orchestration, **model observability**, evaluation, and **retraining pipelines**. Built a conductor-subagent tool-orchestration framework (**42% token reduction**), an LLM evaluation/monitoring framework (RAGAS, BERTScore, latency/accuracy dashboards), and fault-tolerant pipelines (**60% fewer failed runs**). Strong software-engineering fundamentals: version control, testing, code review, documentation.

SKILLS

Tool Integration (MCP-style) • LLM Observability & Telemetry • Model Evaluation (Evals) • Retraining Pipelines • Prompt Engineering • Server Performance • Python • SWE Fundamentals

AI Infrastructure & Tooling: LLM tool integration & orchestration | Multi-Agent Systems | LangChain/LangGraph | Prompt Engineering | Semantic Caching, Model Routing

Observability & Evals: RAGAS, BERTScore, custom metrics | Model Monitoring | Latency/Accuracy Dashboards | Drift Detection | MLflow, Weights & Biases

Languages & Engineering: Python, Java, JavaScript/TypeScript, SQL | FastAPI | Testing, Code Review, Documentation | Git, Jira, Confluence

Infrastructure: Docker, Kubernetes | CI/CD (Jenkins, Azure DevOps) | AWS, Azure, GCP Vertex | PostgreSQL, MongoDB, Redis | Kafka

LLMs & Retrieval: GPT-4o, Claude, Gemini, LLaMA | RAG | FAISS, Pinecone, Weaviate | Fine-tuning (LoRA, QLoRA, PEFT)

PROFESSIONAL EXPERIENCE

Progress Solutions

Jul 2025 - Present

AI Engineer

USA

- Architected production-grade **agentic LLM systems** on a conductor-subagent orchestration framework, decomposing complex objectives into context-scoped subagents that execute autonomously - reducing token consumption by **42%** while sustaining task accuracy across multi-step workflows.
- Engineered high-performance **retrieval-augmented generation (RAG)** pipelines over large-scale healthcare document corpora, combining dense vector retrieval with cross-encoder re-ranking to lift answer relevance and materially reduce hallucination on domain-specific queries.
- Established a structured **LLM evaluation and monitoring framework** (RAGAS, BERTScore, custom domain metrics) paired with latency and accuracy dashboards to benchmark model iterations pre-deployment and surface regressions before release.
- Built fault-tolerant **automation infrastructure** featuring scheduled batch orchestration and exponential-backoff retry logic, improving pipeline reliability and reducing failed production runs by **60%**.
- Optimized inference cost and latency through prompt compression, semantic caching, and model routing, sustaining SLA targets while reducing per-query overhead.

Vanguard

Jan 2024 - Jan 2025

AI Engineer Intern

USA

- Developed and enhanced **AI agents and backend services** for Vanguard's internal AI platform, integrating with production data pipelines and observability tooling across **25+ AI agents** and workflows.
- Built **12 APIs and microservices** powering new AI capabilities, reducing p95 latency from 1.5s to **800ms** and accelerating feature integration by **40%**.
- Optimized LLM prompts and evaluated **GPT-4o, Claude Sonnet, and Gemini** models, improving task success rates by **27%** on internal evaluation datasets.
- Engineered safeguards and content controls for secure, policy-compliant AI responses, reducing unsafe outputs by **42%** while maintaining response quality.
- Partnered with product, platform, and data engineering teams to deliver scalable LLM applications supporting **100,000+ requests daily** across 20+ internal teams.

Kent State University

Oct 2023 - Jan 2024

ML & Gen AI Research Assistant

USA

- Built a **privacy-preserving ML pipeline** on a clinical hospital-readmission dataset (k-anonymity, l-diversity, Laplace differential privacy), reducing re-identification risk from **3.38% to 0.32%** while retaining **99% of baseline predictive performance**.
- Fine-tuned **Qwen3-27B via 4-bit QLoRA** on an NVIDIA H100 using a curated instruction dataset for controllable long-form generation; authored a peer-review-ready IEEE-format paper.

Emerson

Jun 2022 - Apr 2023

ML Trainee

India

- Developed and trained supervised **regression and classification models** for equipment-failure prediction, anomaly detection, and capacity forecasting on operational data.
- Fine-tuned **BERT and RoBERTa** models for entity extraction and text classification, achieving **89% F1-score** on custom NER tasks.

PROJECTS

MangaLens

A shipped Chrome extension (live on the Chrome Web Store) delivering real-time AI translation of manga and webtoons across **29+ sites**, turning a multi-step manual translation chore into a single inline action.

career-ops

An autonomous, AI-driven job-search pipeline that scans listings, scores fit, and generates tailored applications overnight without supervision - compressing hours of repetitive work into an automated workflow.

Privacy-Preserving Clinical ML Pipeline

A framework combining k-anonymity, l-diversity, and differential privacy to let researchers analyze sensitive patient data without exposing identities, quantifying the privacy-utility trade-off of each technique.

EDUCATION

M.S. in Computer Science - Kent State University, OH

2025

GPA: 3.70 / 4.0